Member 5: modelling run on Colab

Runs `python -m src.run_models` (baselines, tuning, comparison, **single** test evaluation, SHAP, secondary target, error analysis), then the rolling-origin step, then builds the slides. Put these two files in Google Drive folder **`MyDrive/m5/`** first:

1. `m5_colab_bundle.zip` (code + weekly counts; made by Claude, see the repo message)
2. `questions_clean_v1.parquet`

Use *Runtime > Change runtime type* and pick a **GPU (T4)** runtime with high-RAM if you can: XGBoost then trains on the GPU automatically (Random Forest, TF-IDF and SVD stay on the CPU; TPUs are not used). Run the cells in order. **The test period is evaluated once**: do not re-run step 4 after it has finished; if Colab disconnects *during* step 4, delete `/content/repo/reports/review1/m5` and run step 4 again.

1. Check the machine

```
import os, subprocess
print('CPU cores:', os.cpu_count())
print(subprocess.run(['free', '-g'], capture_output=True, text=True).stdout)
!nvidia-smi -L || echo 'No GPU: XGBoost will run on the CPU (slower but fine)'
```

```
CPU cores: 2
               total        used        free      shared  buff/cache   available
Mem:              12           3           3           0           6           9
Swap:              0           0           0

GPU 0: Tesla T4 (UUID: GPU-47e5d942-9a60-fd83-07eb-a3f042f8546d)
```

2. Get the files from Drive and install

```
from google.colab import drive
drive.mount('/content/drive')

import shutil, zipfile, pathlib
D = pathlib.Path('/content/drive/MyDrive/m5')
R = pathlib.Path('/content/repo')
shutil.rmtree(R, ignore_errors=True)
zipfile.ZipFile(D / 'm5_colab_bundle.zip').extractall(R)
(R / 'data/processed').mkdir(parents=True, exist_ok=True)
shutil.copy(D / 'questions_clean_v1.parquet', R / 'data/processed/questions_clean_v1.parquet')
print(sorted(p.name for p in R.rglob('*') if p.is_file())[:40])
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_rem
['.gitkeep', '__init__.py', 'aggregate.py', 'build_features.py', 'cleaning.py', 'collect.py', 'collect_questions.py',
```

```
!pip install -q -U shap python-pptx pyarrow xgboost pyyaml
```

```
shell-init: error retrieving current directory: getcwd: cannot access parent directories: No such file or directory
shell-init: error retrieving current directory: getcwd: cannot access parent directories: No such file or directory
The folder you are executing pip from can no longer be found.
```

3. Quick check: load the data and build features (about 1 minute)

Expect **302,026** rows loaded and **293,510** feature rows.

```
%cd /content/repo
!python -W ignore - <<'EOF'
import pandas as pd
from src import build_features as bf
q = pd.read_parquet('data/processed/questions_clean_v1.parquet')
f = bf.build_features(q, pd.read_csv('data/processed/weekly_counts.csv'))
print(len(q), len(f), f.attrs['dropped_no_history'])
```

```
/content/repo
/bin/bash: line 1: warning: here-document at line 1 delimited by end-of-file (wanted `EOF')
302026 293510 8516
```

4. Main run (tuning, comparison, the single test evaluation, SHAP, secondary target, error analysis)

Leaves out the slow rolling-origin step. Takes roughly 30 to 90 minutes on Colab CPU.

```
import os
os.environ['M5_JOBS'] = str(os.cpu_count())
os.environ['M5_DEVICE'] = 'auto'   # GPU for XGBoost if available
!python -W ignore -m src.run_models --no-rolling 2>&1 | tee /content/run_main.log
```

```
[05:57:43] loaded 302,026 rows from data/processed/questions_clean_v1.parquet
[05:57:46] features built: 293,510 rows (8516 dropped: no weekly history yet)
[05:57:47] XGBoost device: cuda (set M5_DEVICE=cpu or cuda to force)
[05:57:47] 30 model inputs, no forbidden columns; unexpected: []
[05:57:48] split:
                rows  target_resolved_7d share  target_answered_7d share
train       248294.0                    0.421                     0.757
validation   28132.0                    0.341                     0.702
test         16695.0                    0.297                     0.714
[05:58:24] feature gain (validation):
                            f1  pr_auc  roc_auc
model
structured only          0.5176  0.4442   0.6128
structured + title text  0.5250  0.4707   0.6361
[06:33:26] random forest: best CV PR-AUC 0.5252 with {'max_depth': 16, 'min_samples_leaf': 24, 'n_estimators': 472} (
[06:34:21] xgboost: best CV PR-AUC 0.5170 with {'learning_rate': 0.029447905545150106, 'max_depth': 5, 'n_estimators'
[07:00:08] validation comparison:
                    precision  recall      f1  pr_auc  roc_auc  capture_rate_20
model
majority class          0.0000  0.0000  0.0000  0.3405   0.5000           0.2012
logistic regression     0.3887  0.8085  0.5250  0.4707   0.6361           0.2399
random forest           0.3861  0.8130  0.5236  0.4734   0.6382           0.2378
xgboost                 0.4003  0.7819  0.5295  0.4821   0.6448           0.2408
[07:00:09] chosen: xgboost (threshold 0.367)
[07:00:10] TEST (once):
model               xgboost
n                16695.0000
positive_rate        0.2969
threshold            0.3669
precision            0.3626
recall               0.7712
f1                   0.4933
pr_auc               0.4315
roc_auc              0.6548
capture_rate_20      0.2415
[07:00:16] top SHAP features: lagged_tag_volume, account_age_days, code_ratio, n_code_blocks, title_tfidf_svd_5
[07:00:46] secondary target (test):
                                    precision  recall      f1  pr_auc  roc_auc
model
main: accepted within 7 days           0.3626  0.7712  0.4933  0.4315   0.6548
secondary: any answer within 7 days    0.7142  0.9999  0.8333  0.8300   0.6670
[07:00:46] done: results in reports/review1/m5/results.json
```

Save a copy to Drive right away, so a disconnect cannot lose it.

```
import shutil
shutil.make_archive('/content/drive/MyDrive/m5/m5_results_main', 'zip', '/content/repo/reports/review1/m5')
print('saved m5_results_main.zip to Drive')
```

```
saved m5_results_main.zip to Drive
```

5. Rolling-origin evaluation (slow; 11 half-year windows, each refits the model twice)

Uses the saved settings of step 4. It does not touch the test period.

```
!python -W ignore -m src.run_models --rolling-only 2>&1 | tee /content/run_rolling.log
```

```
[07:00:51] loaded 302,026 rows from data/processed/questions_clean_v1.parquet
[07:00:55] features built: 293,510 rows (8516 dropped: no weekly history yet)
[07:09:54] rolling-origin: 11 windows (540s); mean PR-AUC pre 0.559, post 0.467
[07:09:56] rolling results added to reports/review1/m5/results.json
```

6. Slides and findings text

```
!python -m src.make_slides
```

```
wrote reports/review1/m5_slides.pptx
wrote reports/review1/m5_findings.md
```

7. Copy everything back to Drive

```
import shutil
shutil.make_archive('/content/drive/MyDrive/m5/m5_results_all', 'zip', '/content/repo/reports/review1/m5')
for f in ('m5_slides.pptx', 'm5_findings.md'):
    shutil.copy('/content/repo/reports/review1/' + f, '/content/drive/MyDrive/m5/' + f)
print('Done: m5_results_all.zip, m5_slides.pptx, m5_findings.md are in Drive/m5')
```

```
Done: m5_results_all.zip, m5_slides.pptx, m5_findings.md are in Drive/m5
```